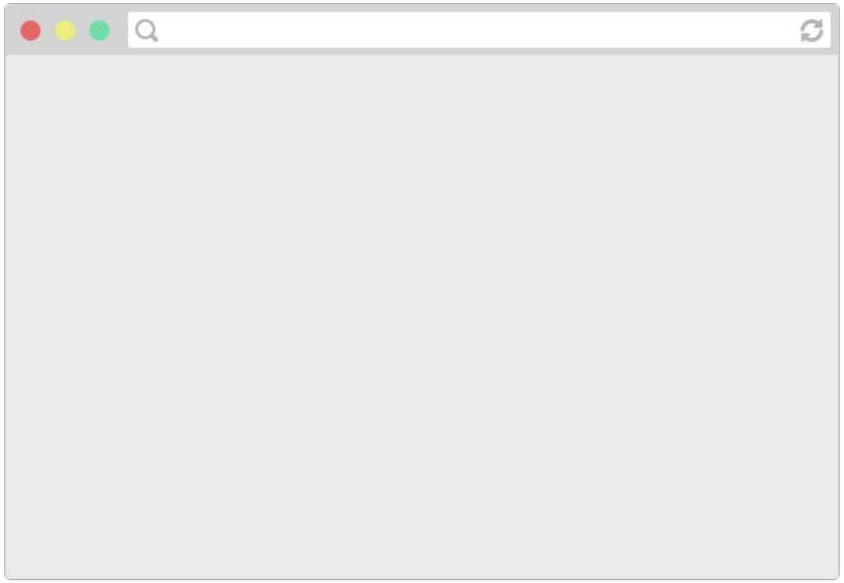
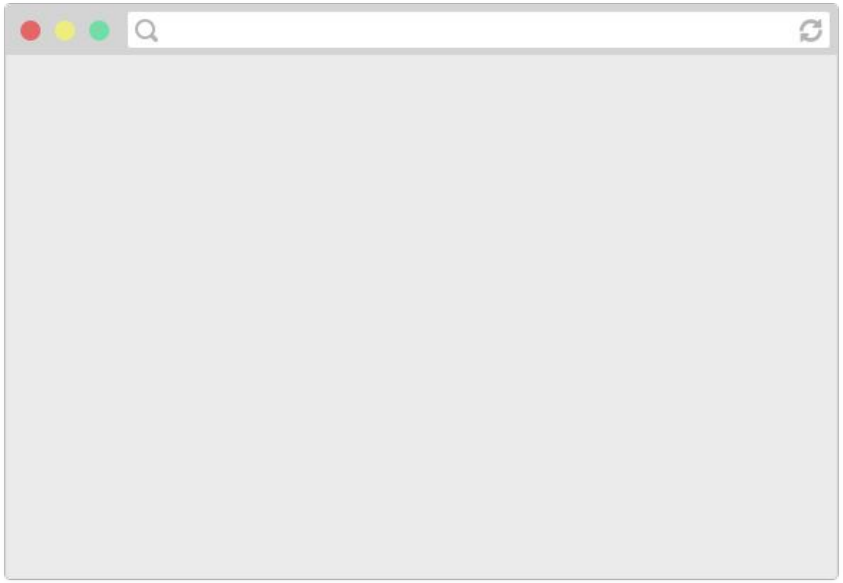
Browsers!
The Internet!
The web!

How do web pages work?



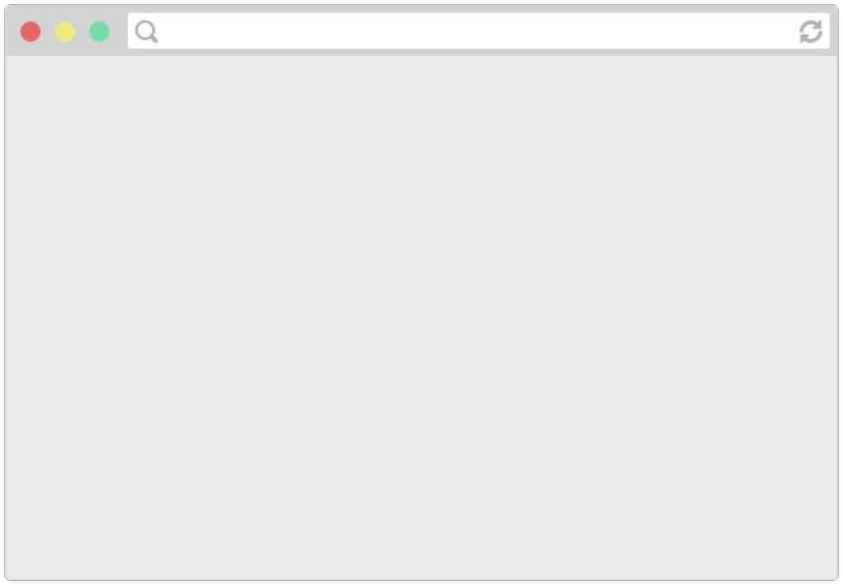
Browsers are applications that can display web pages.
E.g. Chrome, Firefox, Safari, Internet Explorer, Edge, etc.

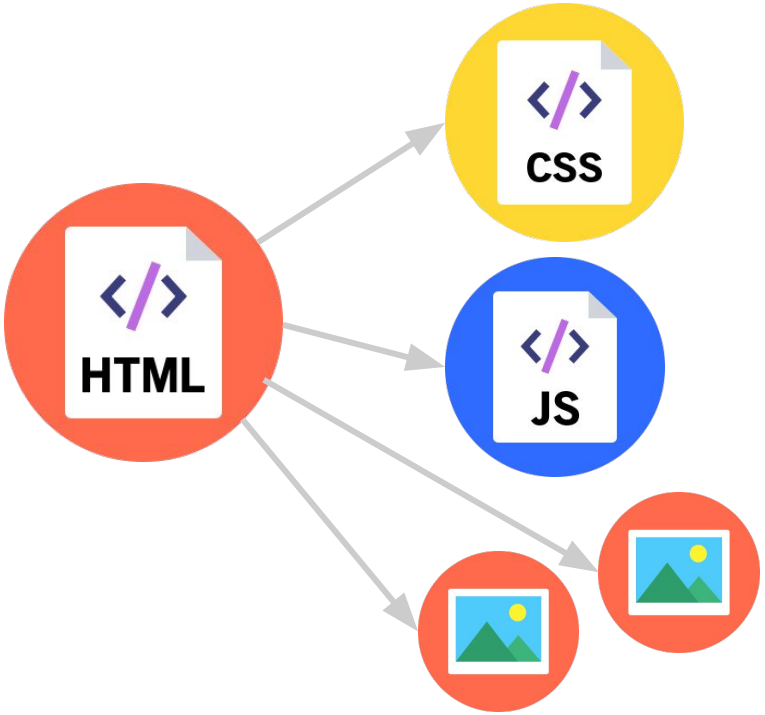
How do web pages work?



Web pages are written in a markup language called **HTML**, so browsers display a web page by reading and interpreting its HTML.

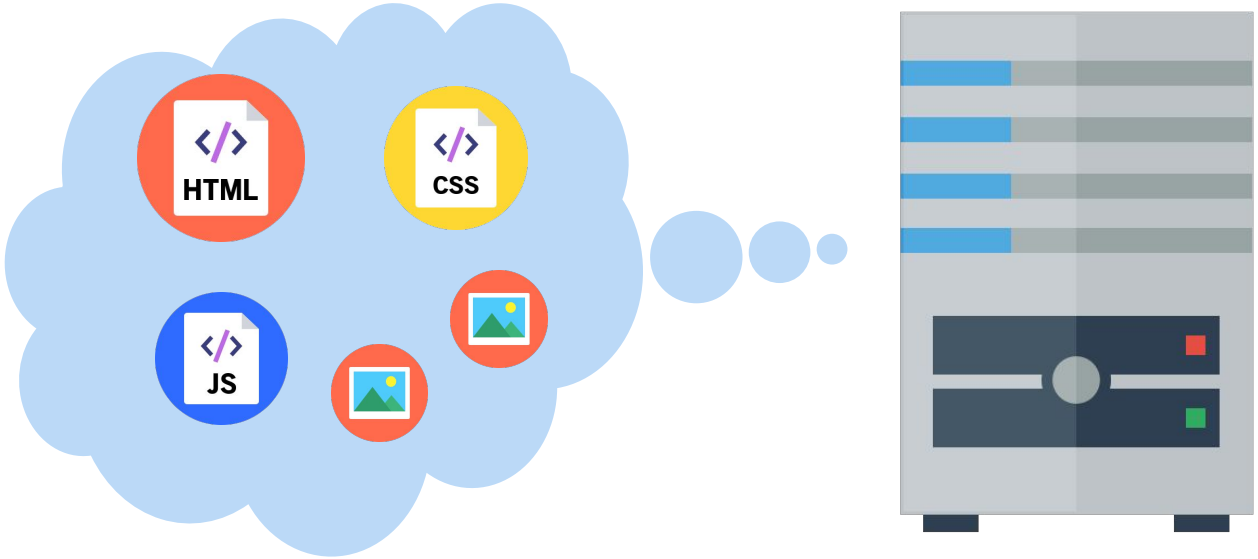
How do web pages work?





The HTML file might link to other resources, like images, videos, as well as **JavaScript** and **CSS** (stylesheet) files, which the browser then also loads.

How do web pages work?

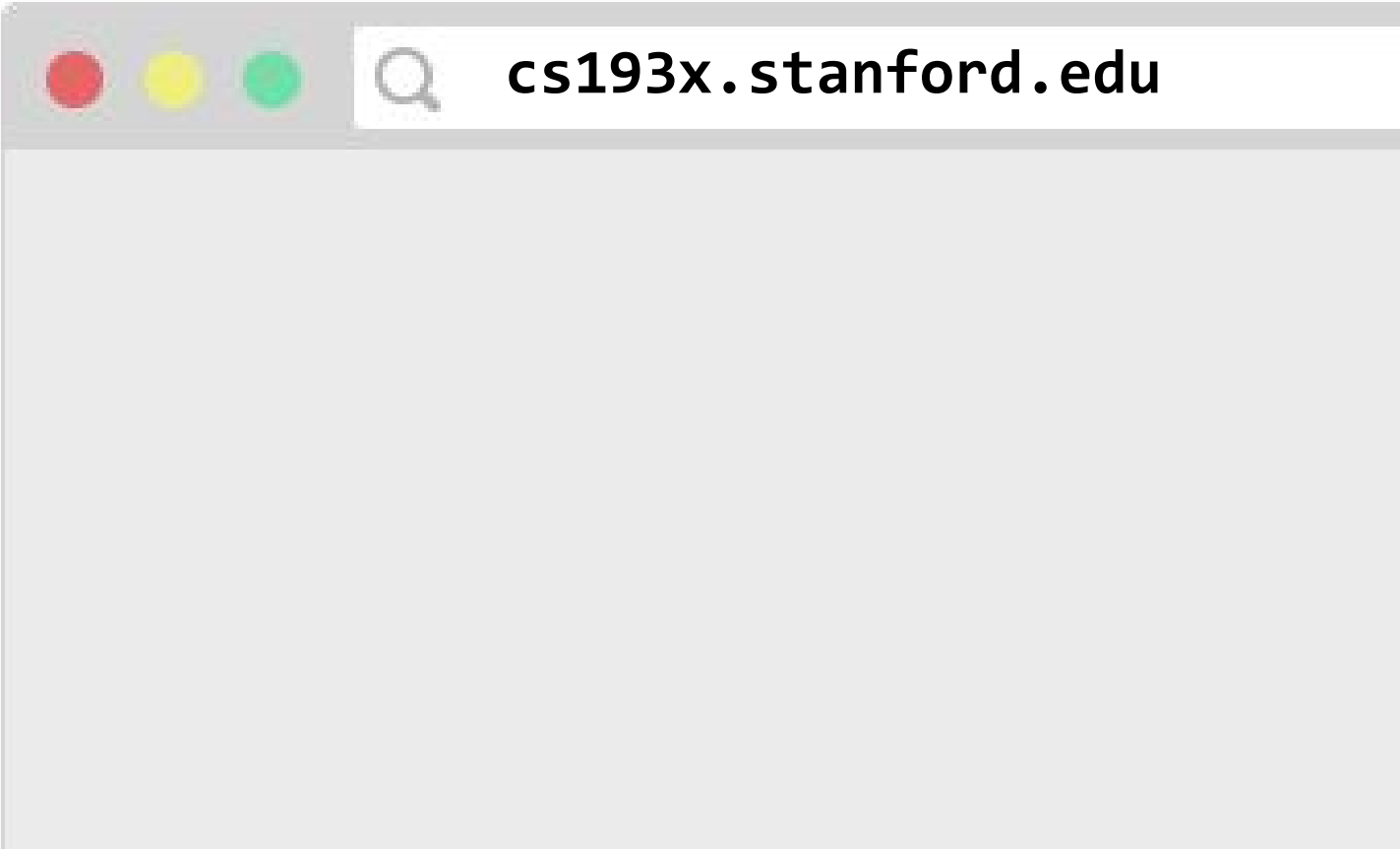


A **web server** is a program running on a computer that delivers web pages in response to requests.

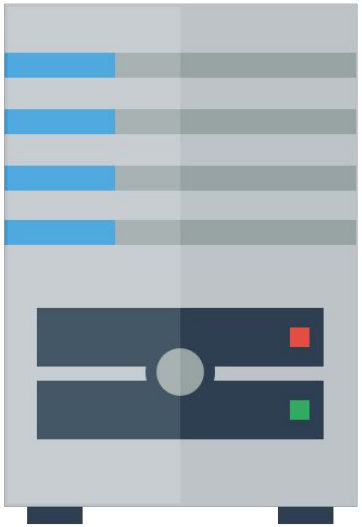
It either stores or generates the web page returned.

How do web pages work?

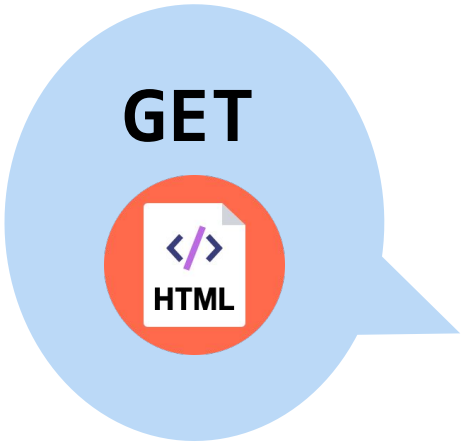
1. You type in a URL, which is the address of the HTML file on the internet.

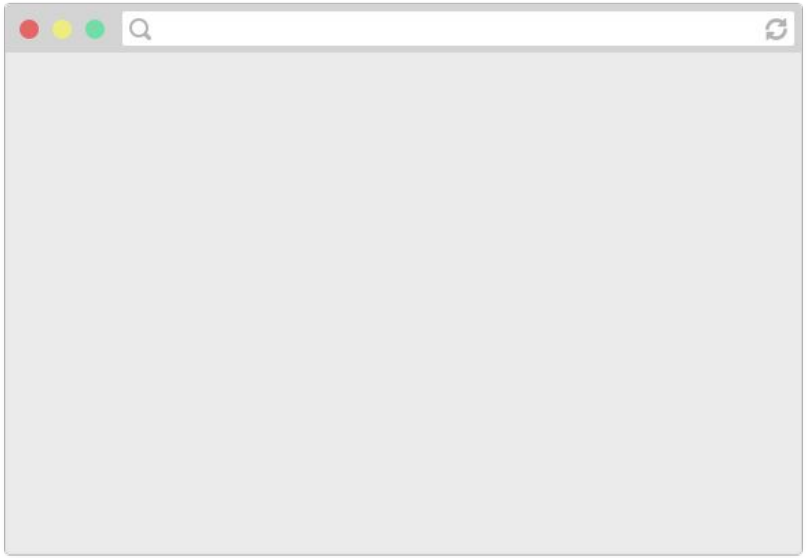


How do web pages work?

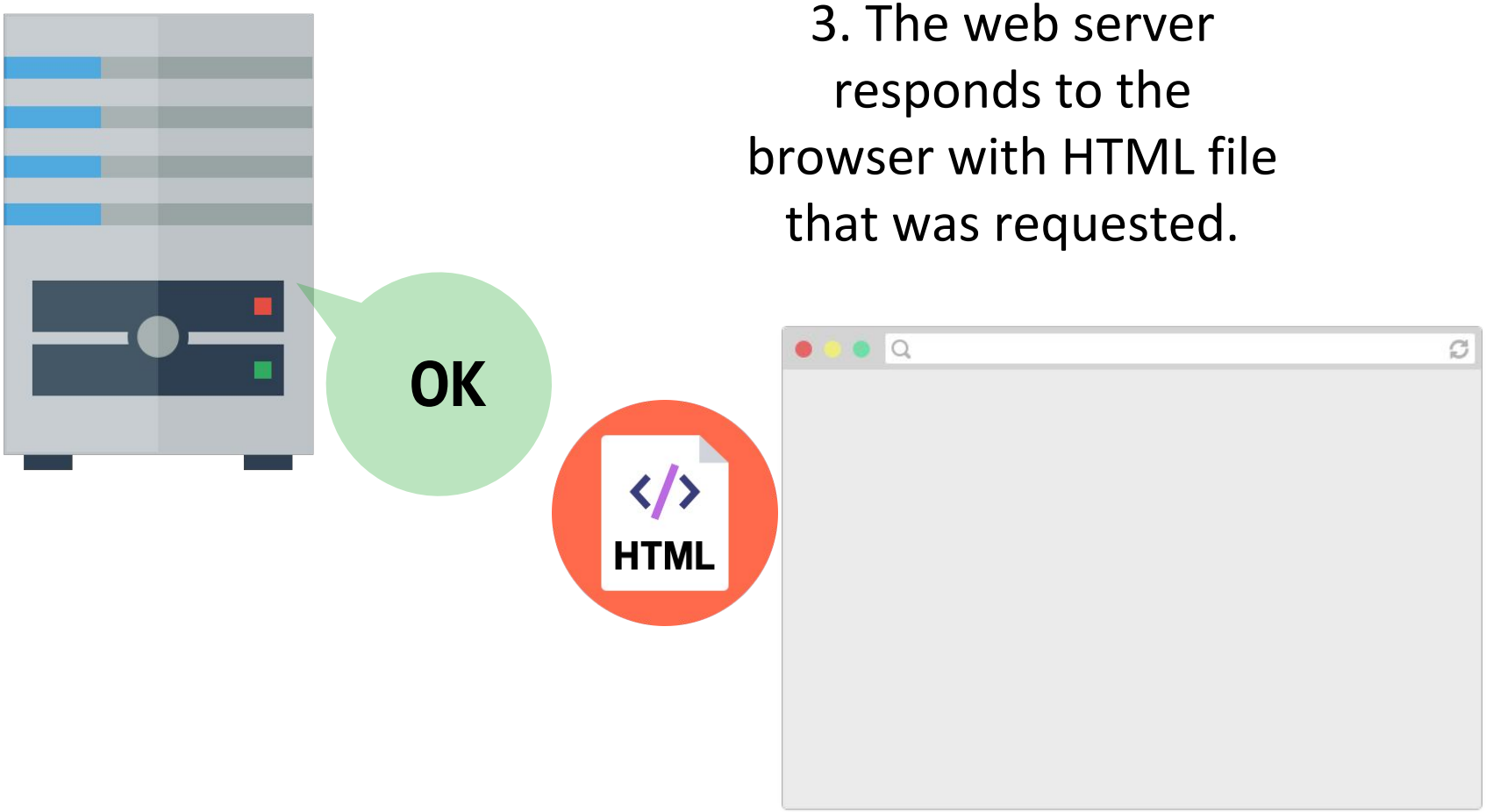


2. The browser asks the web server that hosts the document to send that document.





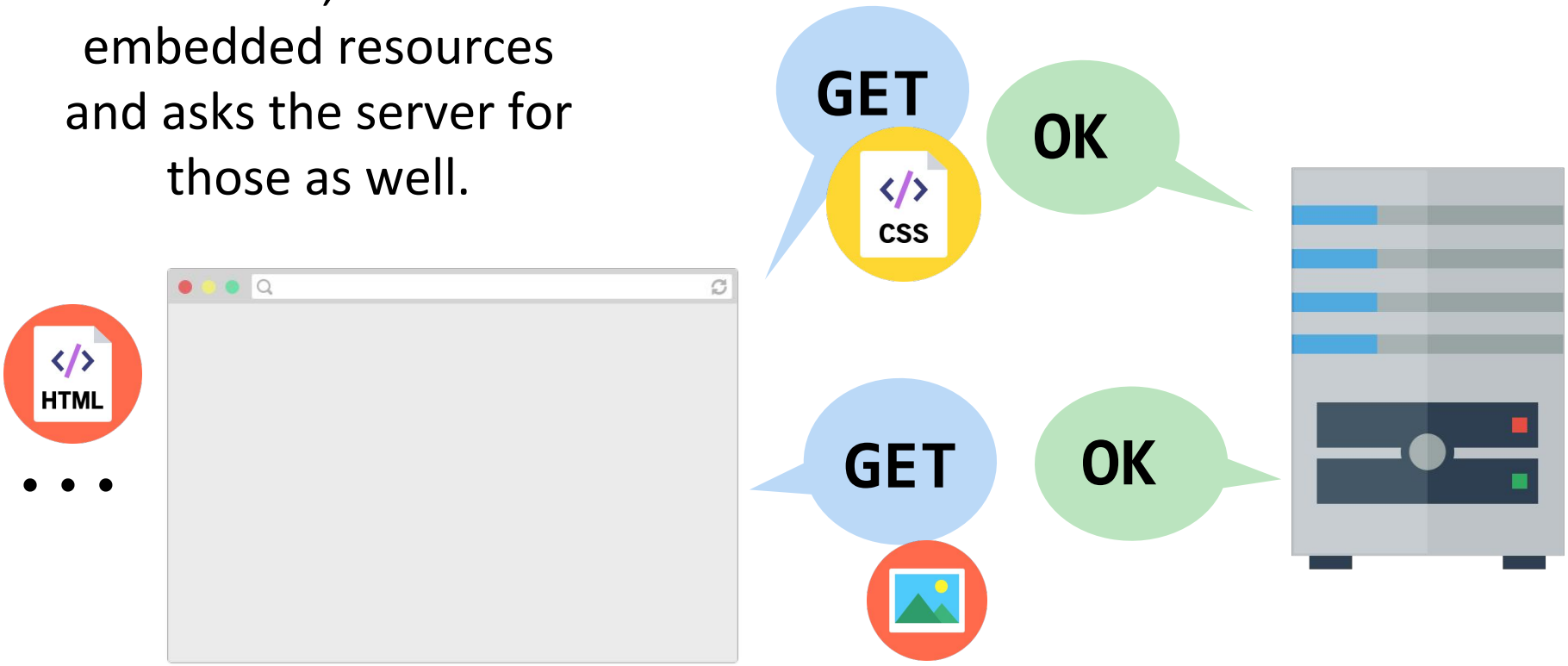
How do web pages work?



3. The web server responds to the browser with HTML file that was requested.

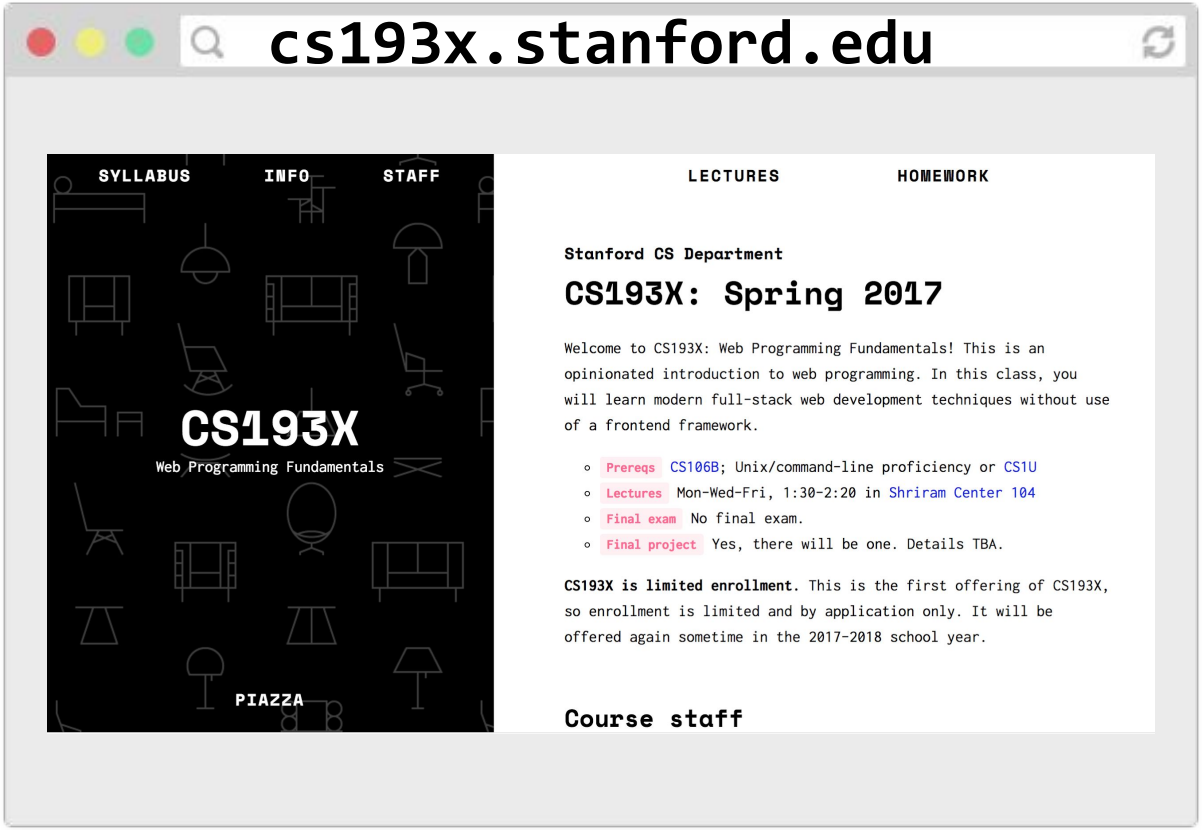
How do web pages work?

4. The browser reads the HTML, sees the embedded resources and asks the server for those as well.



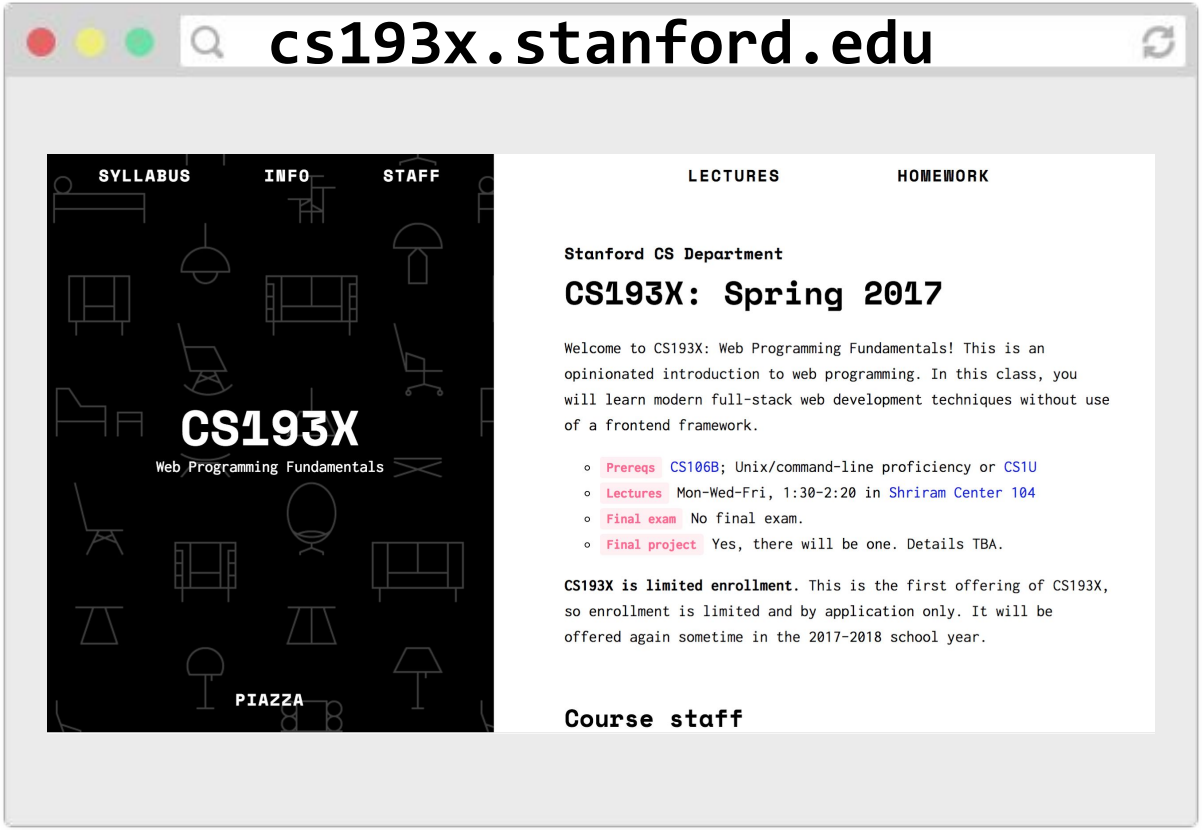
How do web pages work?

5. The web page is loaded when all the resources are fetched and displayed.



P.S.

(That was obviously very hand-wavy. We'll get more detailed when we talk about servers later in the quarter.)



HTML and CSS

HTML and CSS strategy

Assumption: Most people have cursory familiarity with HTML and CSS. Therefore we will:

- **Speed through** the obvious stuff
- **Skip** self-explanatory syntax
- **Skip** the parts you can look up easily through Google

✨ **Therefore, be aggressive with questions!** ✨

What is HTML?

HTML (**H**ypertext **M**arkup **L**anguage)

- Describes the **content** and **structure** of a web page; not a programming language.
- Made up of building blocks called **elements**.

```
<p>
  HTML is <em>awesome!!!</em>
  <img src="puppy.png" />
</p>
```

Basic HTML page structure

(i.e. copy/paste boilerplate)

```
<!DOCTYPE html>
<html>
  <head>
   <title>CS 193X</title>
  </head>

  <body>
   ... contents of the page...
  </body>
</html>
```

Saved in a *filename*`.html` file.

Basic HTML page structure

(i.e. copy/paste boilerplate)

Metadata that doesn't appear in the viewport of the browser

Contents that render in the viewport of the browser

```
<!DOCTYPE html>
<html>
  <head>
    <title>CS 193X</title>
  </head>

  <body>
    ... contents of the page...
  </body>
</html>
```

E.g. **`<title>`** shows up as the name of the tab

HTML elements

```
<p>
  HTML is <em>awesome!!!</em>
  <img src="puppy.png" />
</p>
```

- An element usually has start and ending tags (**`<p>`** and **`</p>`**)
  - **content**: stuff in between start and end tags
- An element can be self-closing (**`img`**)
- An element can have attributes (**`src="puppy.jpg"`**)
- Elements can contain other elements (**`p`** contains **`em`** and **`img`**)

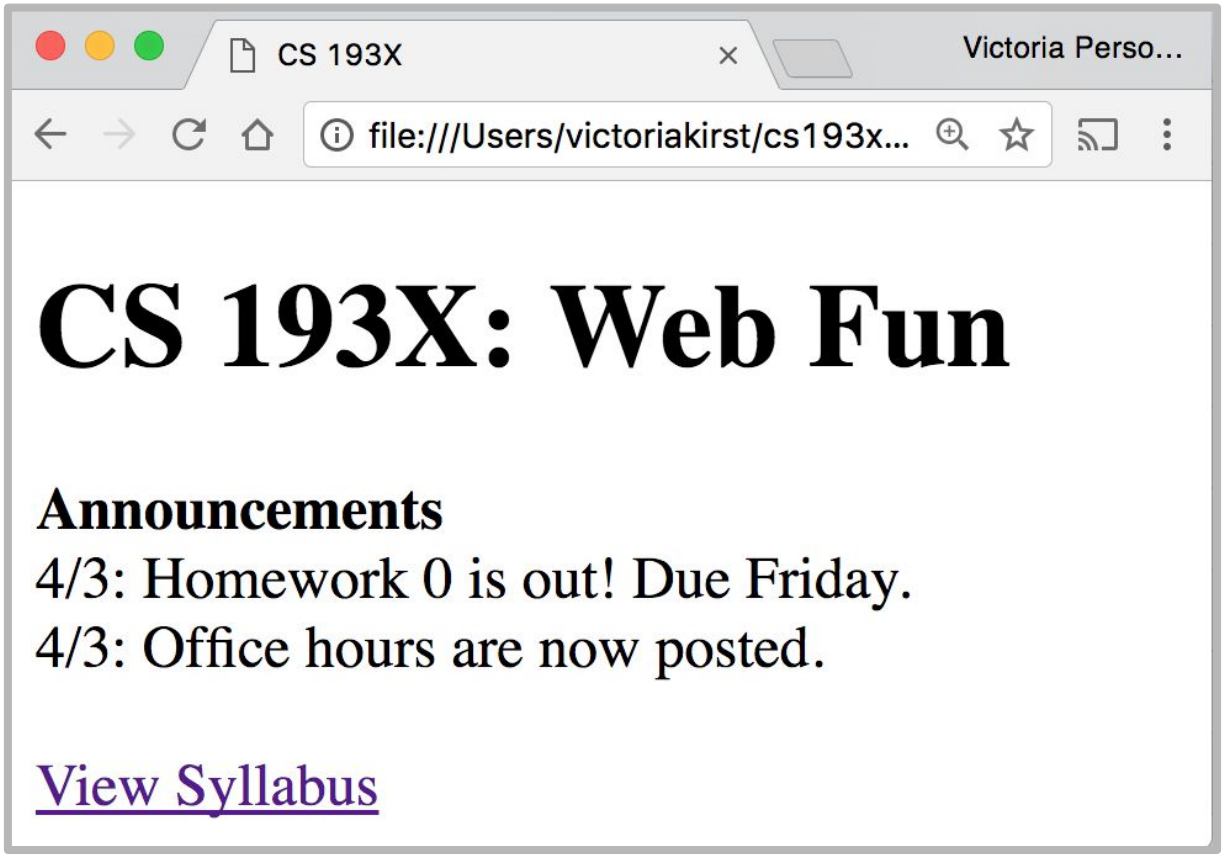
Some HTML elements

(to place within **`<body>`**)

Top-level heading **`h1, h2, … h6`**	`<h1>Moby Dick</h1>`
Paragraph	`<p>Call me Ishmael.</p>`
Line break	`since feeling is first who pays any attention`
Image	`<img src="cover.png" />`
Link	`<a href="google.com">click here!</a>`
Strong (bold)	`<strong>Be BOLD</strong>`
Emphasis (italic)	`He's my <em>brother</em> and all`

Exercise: Course web page

Let's write some HTML to make the following page:



Exercise: Course web page

HTML boilerplate

```
<!DOCTYPE html>
<html>
  <head>
    <title>CS 193X</title>
  </head>

  <body>
   ...
  </body>
</html>
```

Plaintext contents of the page

```
CS 193X: Web Fun

Announcements
4/3: Homework 0 is out! Due
Friday.
4/3: Office hours are now
posted.

View Syllabus
```

JSBin

Solution

```
<!DOCTYPE html>
<html>
  <head>
    <title>CS 193X</title>
  </head>
  <body>
    <h1>CS 193X: Web Fun</h1>
    <strong>Announcements</strong><br/>
    4/3: Homework 0 is out!<br/>
    4/3: Office hours are now posted.<br/>
    <br/>
    <a href="http://cs193x.stanford.edu/syllabus">
      View Syllabus
    </a>
  </body>
</html>
```

That was weird

- We saw that HTML whitespace collapses into one space…

```
<h1>CS 193X: Web Fun</h1>
<strong>Announcements</strong><br/>
4/3: Homework 0 is out!<br/>
```

- Except weirdly the **`<h1>`** heading was on a line of its own, and **`<strong>`** was not.

Hmmm… strange…
Oh well, it works! Let's move on!!!

CSS

CSS

CSS: **C**ascading **S**tyle Sheets

- Describes the **appearance** and **layout** of a web page
- Composed of CSS **rules**, which define sets of styles

```
selector {
  property: value;
}
```

CSS

A CSS file is composed of **style rules**:

```
selector {
  property: value;
}
```

selector: Specifies the HTML element(s) to style.
property: The name of the CSS style.
value: The value for the CSS style.

Saved in a *filename*`.css` file.

CSS

```
// NOT REAL CSS
fork {
  color: gold;
}
```

"All forks on the table should be gold"

CSS

```
p {
  color: blue;
  font-weight: bold;
}
```

"All **<p>** elements on the page should be blue and bold"

Linking CSS in HTML

(i.e. copy/paste boilerplate)

```
<!DOCTYPE html>
<html>
  <head>
    <title>CS 193X</title>
    <link rel="stylesheet" href="filename.css" />
  </head>

  <body>
   ... contents of the page...
  </body>
</html>
```

Some CSS properties

There are over 500 CSS properties! Here are a few:

Font face (mdn)	`font-family: Helvetica;`
Font color (mdn)	`color: gray;`
Background color (mdn)	`background-color: red;`
Border (mdn)	`border: 3px solid green;`
Text alignment (mdn)	`text-align: center;`

Aside: Mozilla Developer Network (MDN) is the best reference for HTML elements and CSS properties

- The actual W3 spec is very hard to read (meant for browser developers, not web developers)

Main ways to define CSS colors:

140 predefined names (list)

```
color: black;
```

rgb() and rgba()

```
color: rgb(34, 12, 64);
color: rgba(0, 0, 0, 0.5);
```

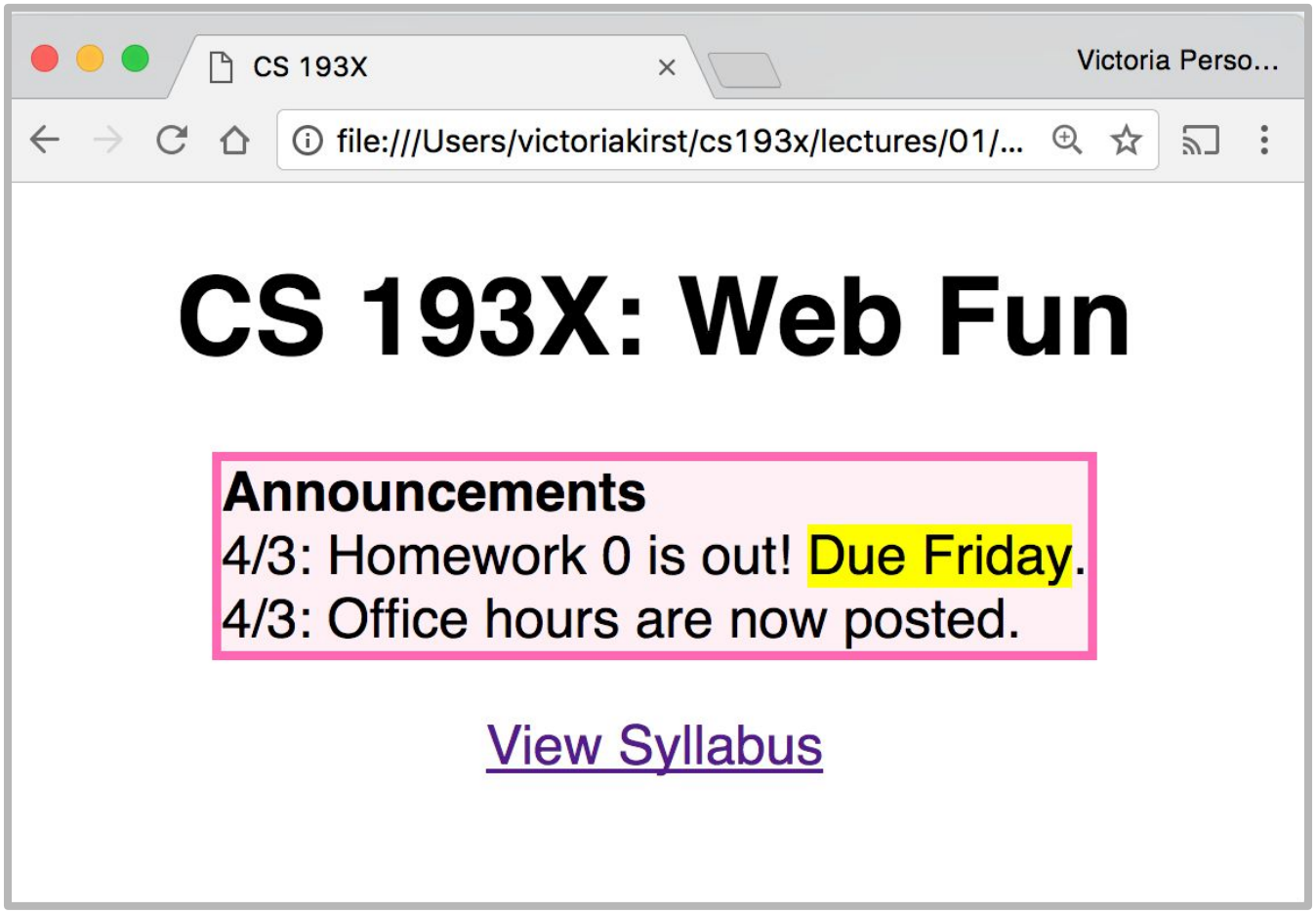
Hex values

```
color: #00ff00;
color: #0f0;
color: #00ff0080;
```

- The "a" stands for **alpha channel** and is a **transparency** value
- Generally prefer more descriptive over less:
  1. Predefined name
  2. rgb / rgba
  3. Hex

Exercise: Course web page

Let's write some CSS to style our page:



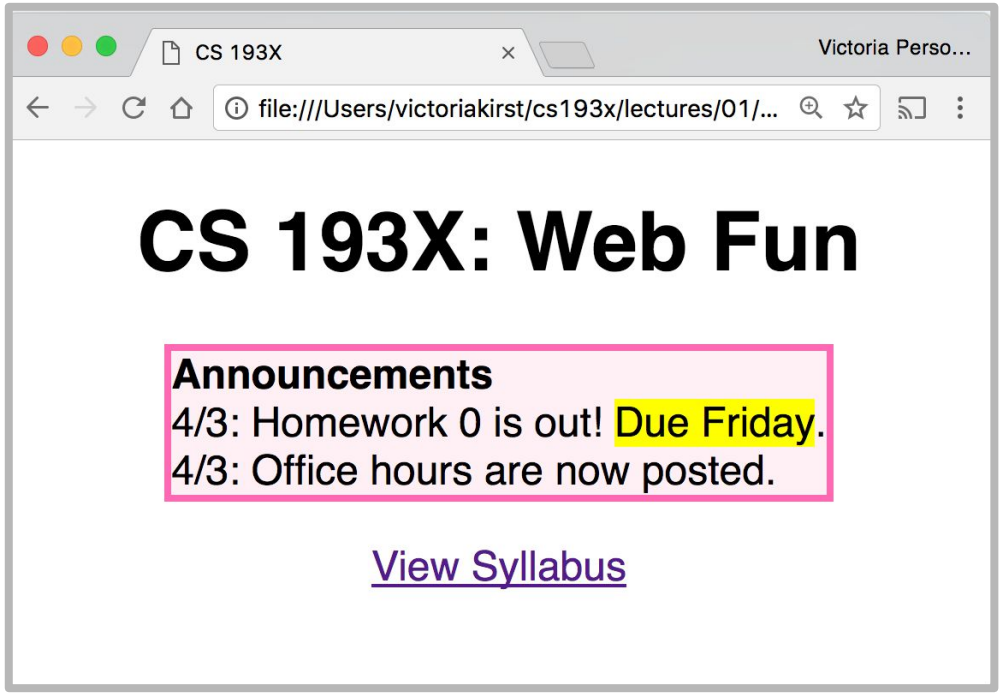
Exercise: Course web page

Let's write some CSS to style our page:

Font face: Helvetica

Border: `hotpink 3px`
Background color: `lavenderblush`
Highlight: `yellow`

- Box is **centered**
- Header and link are **centered**
- Box contents are **left-aligned**



JSBin

CSS exercise debrief

Some **key techniques:**

- Add invisible containers in HTML to select groups of elements in CSS.
- Apply styles to parent / ancestor element to style parent and all its children. (Will talk more about this later.)

But we encountered **more weirdness**...

- Couldn't set `text-align: center;` to the `<a>` or `<strong>` tags directly, but could center `<p>` and `<h1>`
- Had to set a `width` on the box to make it hug the text … any other way to do this?
- How to center the box?! How do you highlight?!

Q: Why is HTML/CSS so bizarre??

A: There is one crucial set of rules we haven't learned yet...

block vs **inline** display

Next time!

Homework 0 is
out now, due this Friday
April 7

Overflow slides

Q: Why is HTML/CSS so bizarre??

A: There is one crucial set of rules we haven't learned yet...

block vs **inline** display

What is HTML?

HTML (**H**yper**t**ext **M**arkup **L**anguage)

- Describes the **content** and **structure** of a web page
- Made up of building blocks called **elements**.

```
<p>
  HTML is <em>awesome!!!</em>
  <img src="puppy.png" />
</p>
```

And there are 3 basic types.

Types of HTML elements

Each HTML element is categorized by the HTML spec into one of three-ish categories:

1. **block:** large blocks of content, has height and width
   `<p>`, **`<h1>`**, `<blockquote>`, `<ol>`, `<ul>`, `<table>`
2. **inline:** small amount of content, no height or width
   `<a>`, `<em>`, **`<strong>`**,`<br>`
   a. **inline block:** inline content with height and width
      `<img>`
3. metadata: information about the page, usually not visible
   `<title>`, `<meta>`

Block elements

Examples:

`<p>, <h1>, <blockquote>, <ol>, <ul>, <table>`

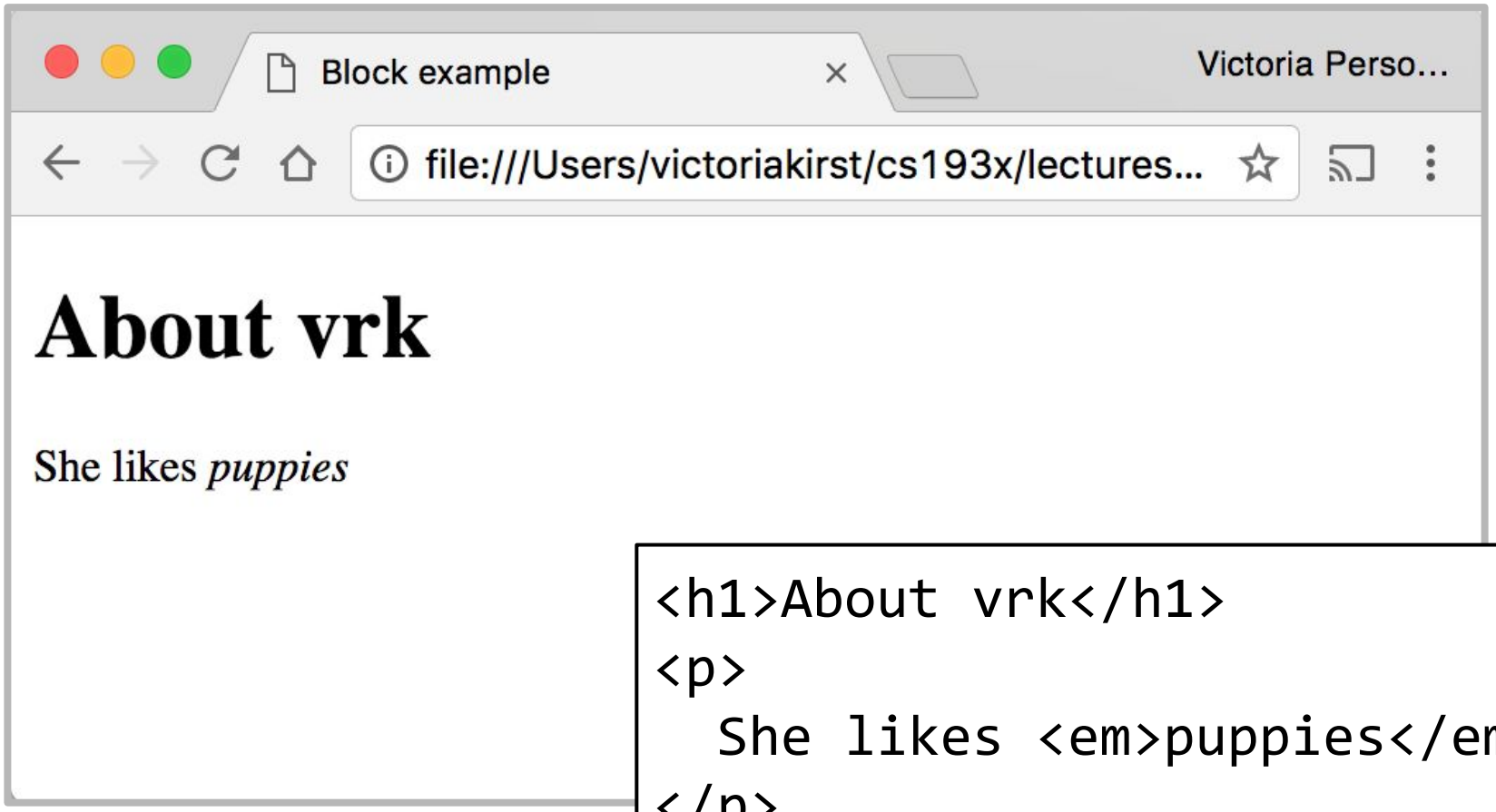
- Take up the full width of the page (**flows top to bottom**)
- Have a `height` and `width`
- Can have block or inline elements as children

Example: Block

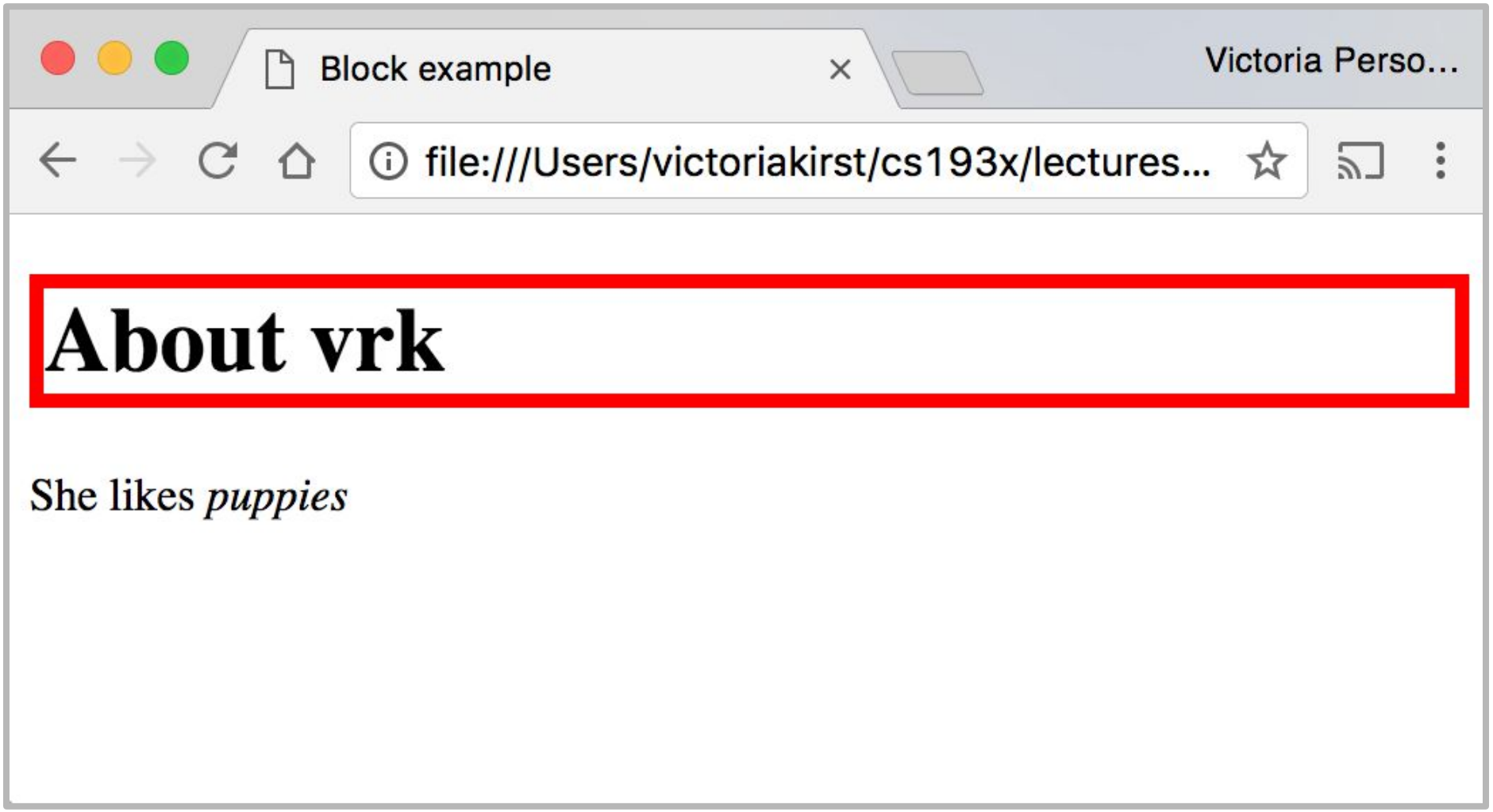
```
<h1>About vrk</h1>
<p>
  She likes <em>puppies</em>
</p>
```

Q: What does this look like in the browser?

```
h1 {
  border: 5px solid red;
}
```




```
<h1>About vrk</h1>
<p>
  She likes <em>puppies</em>
</p>
```


Block example
Victoria Perso...
file:///Users/victoriakirst/cs193x/lectures...
About vrk
She likes puppies

Block-level:

extends the full width of the page

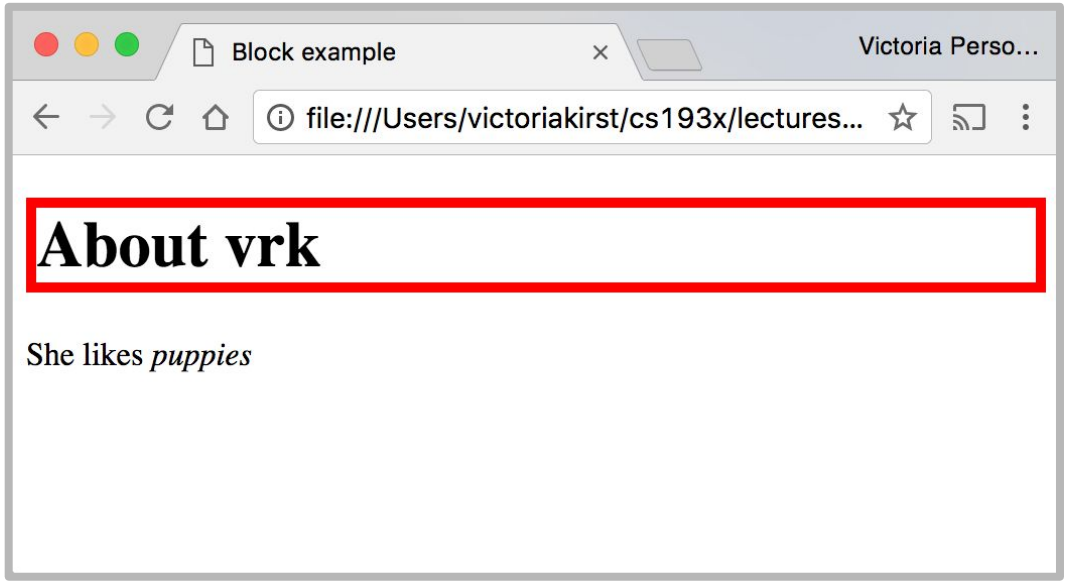
```
h1 {
  border: 5px solid red;
}
```

```
<h1>About vrk</h1>
<p>
  She likes <em>puppies</em>
</p>
```

<h1> is block-level, so it extends the full width of the page by default

Note how block-level elements (**h1**, **p**) flow top to bottom

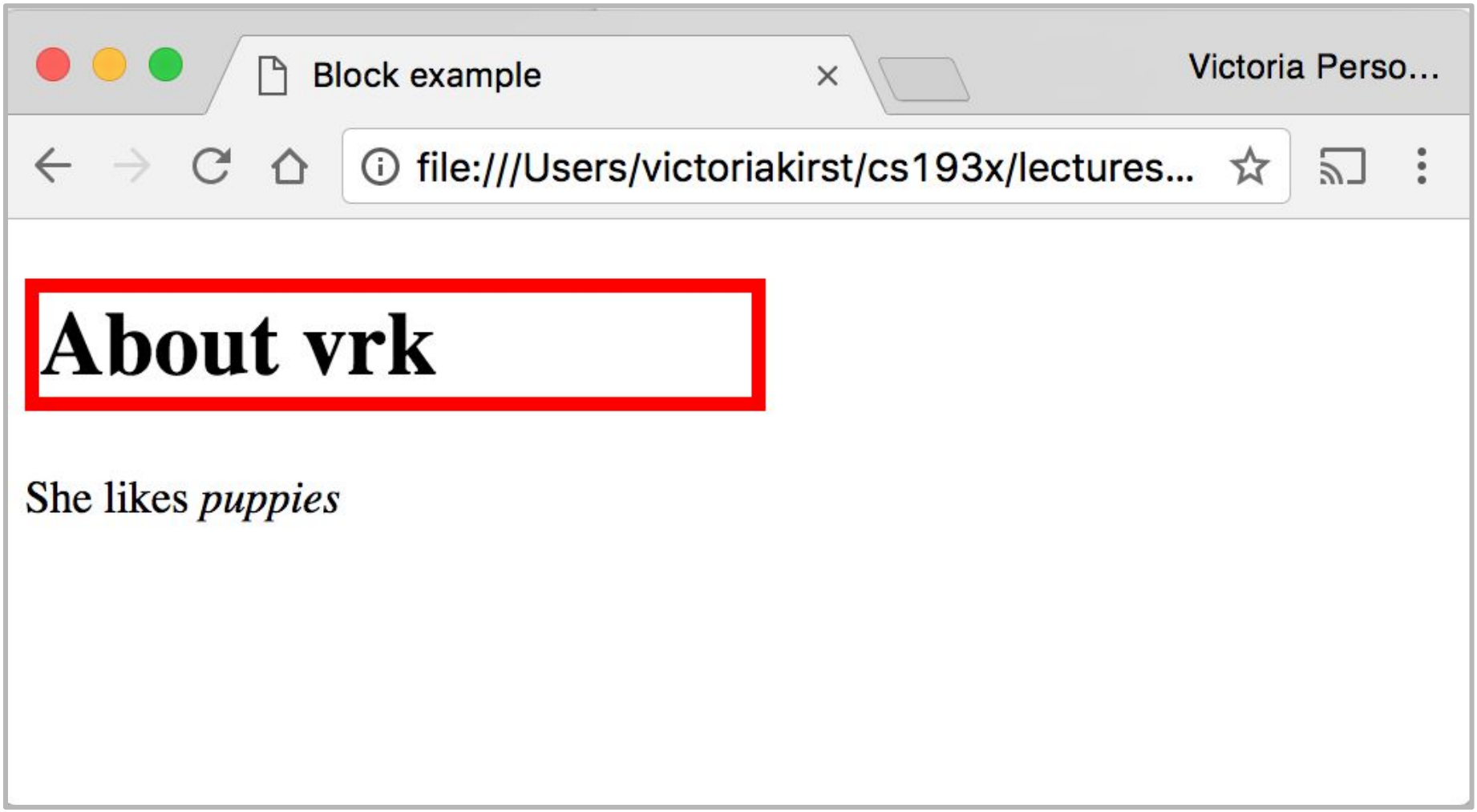
See: [JSBin]



Q: What does this look like in the browser?

```
h1 {
  border: 5px solid red;
  width: 50%;
}
```

```
<h1>About vrk</h1>
<p>
  She likes <em>puppies</em>
</p>
```


Block example
Victoria Perso...
file:///Users/victoriakirst/cs193x/lectures...
About vrk
She likes puppies

Block-level

width can be modified

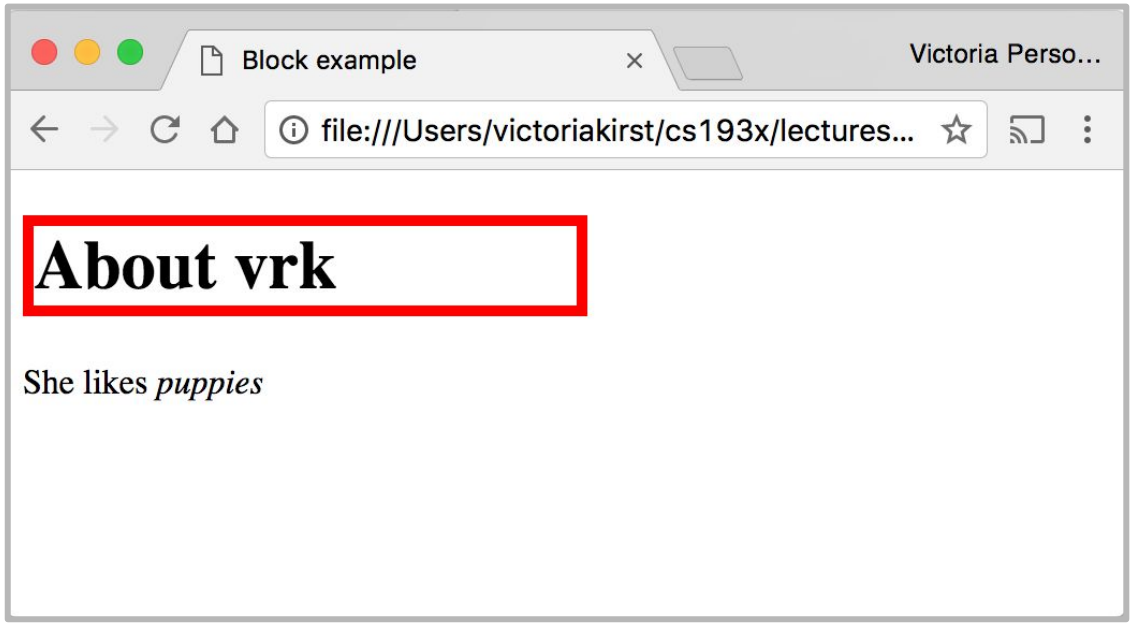
```
h1 {
  border: 5px solid red;
  width: 50%;
}
```

```
<h1>About vrk</h1>
<p>
  She likes <em>puppies</em>
</p>
```

`<h1>` is block-level, so its **`width`** **can** be modified

Block-level elements still flow top to bottom

See: [JSBin]

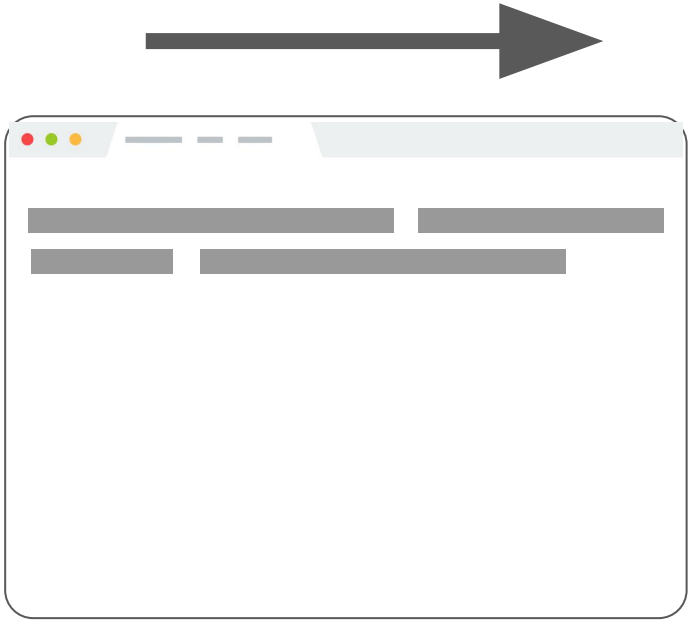


Inline elements

Examples:

`<a>, <em>, <strong>, <br>`

- Take up only as much width as needed (flows left to right)
- **Cannot** have `height` and `width`
- **Cannot** have a block element child
- **Cannot** be positioned (i.e. CSS properties like `float` and `position` do not apply to inline elements)
    - Must position **its containing block element** instead

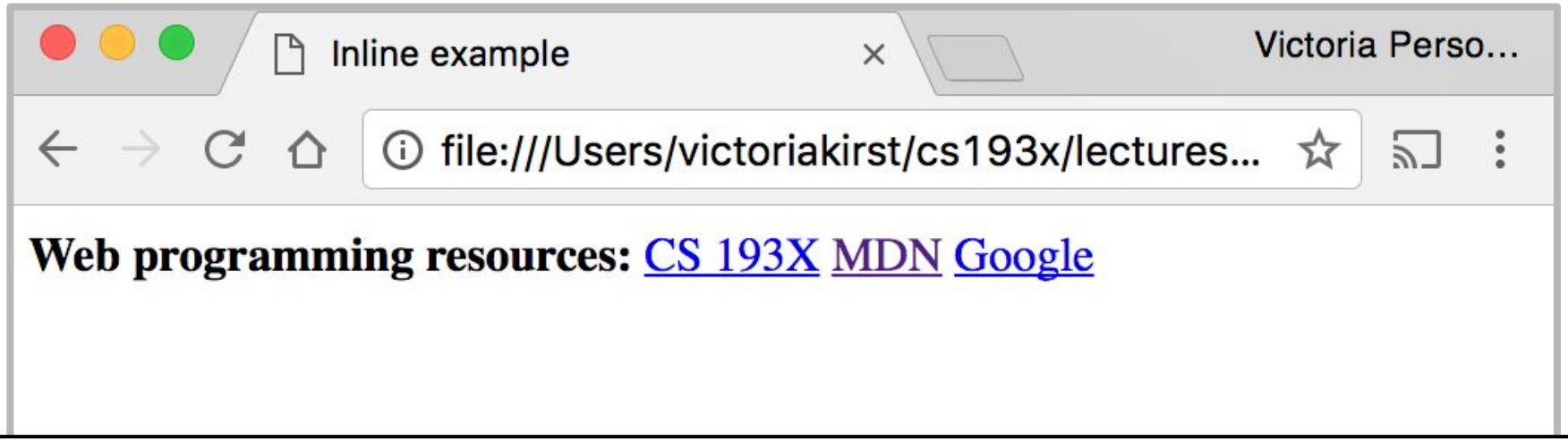


Example: Inline

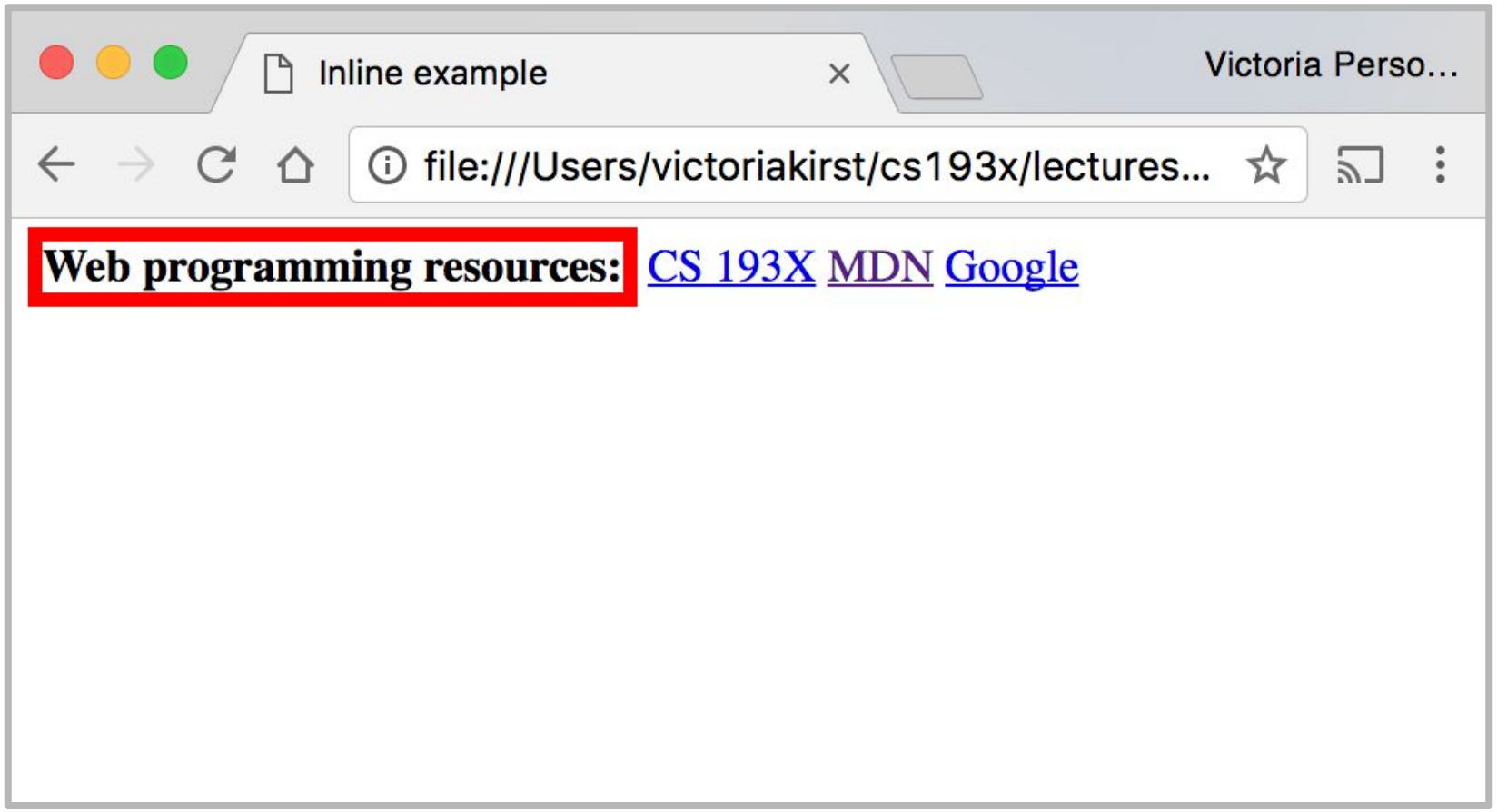
```
<strong>Web programming resources:</strong>
<a href="http://cs193x.stanford.edu">CS 193X</a>
<a href="https://developer.mozilla.org/en-US/">MDN</a>
<a href="http://google.com">Google</a>
```

Q: What does this look like in the browser?

```
strong {
  border: 5px solid red;
  width: 1000px;
}
```




```
<strong>Web programming resources:</strong>
<a href="http://cs193x.stanford.edu">CS 193X</a>
<a href="https://developer.mozilla.org/en-US/">MDN</a>
<a href="http://google.com">Google</a>
```

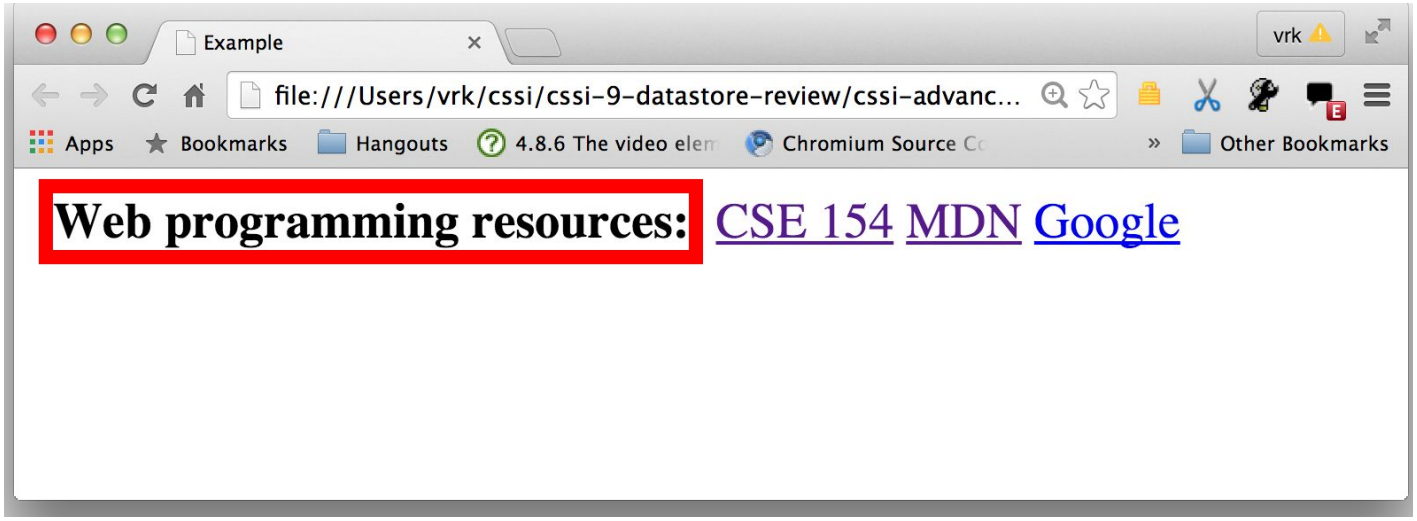

Inline example
Victoria Perso...
file:///Users/victoriakirst/cs193x/lectures...
Web programming resources:
CS 193X
MDN
Google

Inline elements ignore `width`

width cannot be modified

```
strong {
  border: 5px solid red;
  width: 1000px;
  /* Will not work; strong is
     inline! */
}
```

```
<strong>Web programming reso
<a href="http://cs193x.stanf
<a href="https://developer.m
<a href="http://google.com">
```

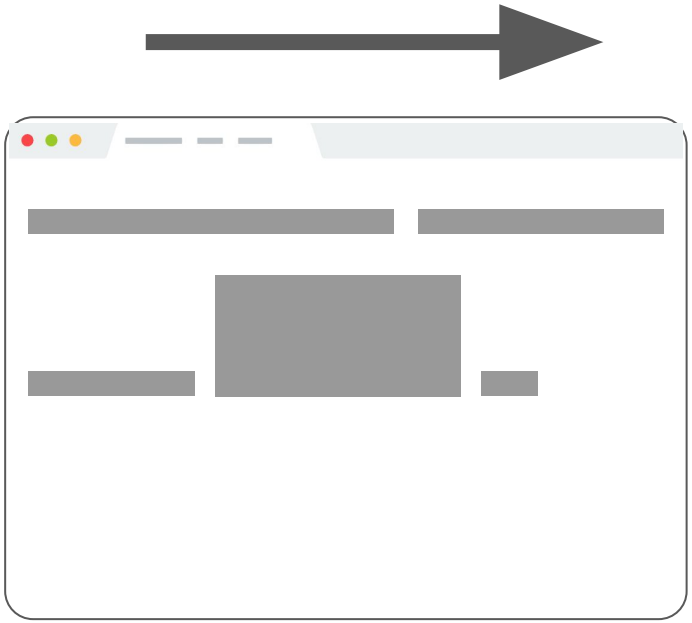


Cannot set **`width`** on inline element, so it is ignored (JSBin)

`inline-block`

Examples: `<img>`, any element with `display: inline-block;`

- Take up only as much width as needed (flows left to right)
- **Can** have `height` and `width`
- **Can** have a block element as a child
- **Can** be positioned (i.e. CSS properties like `float` and `position` apply)



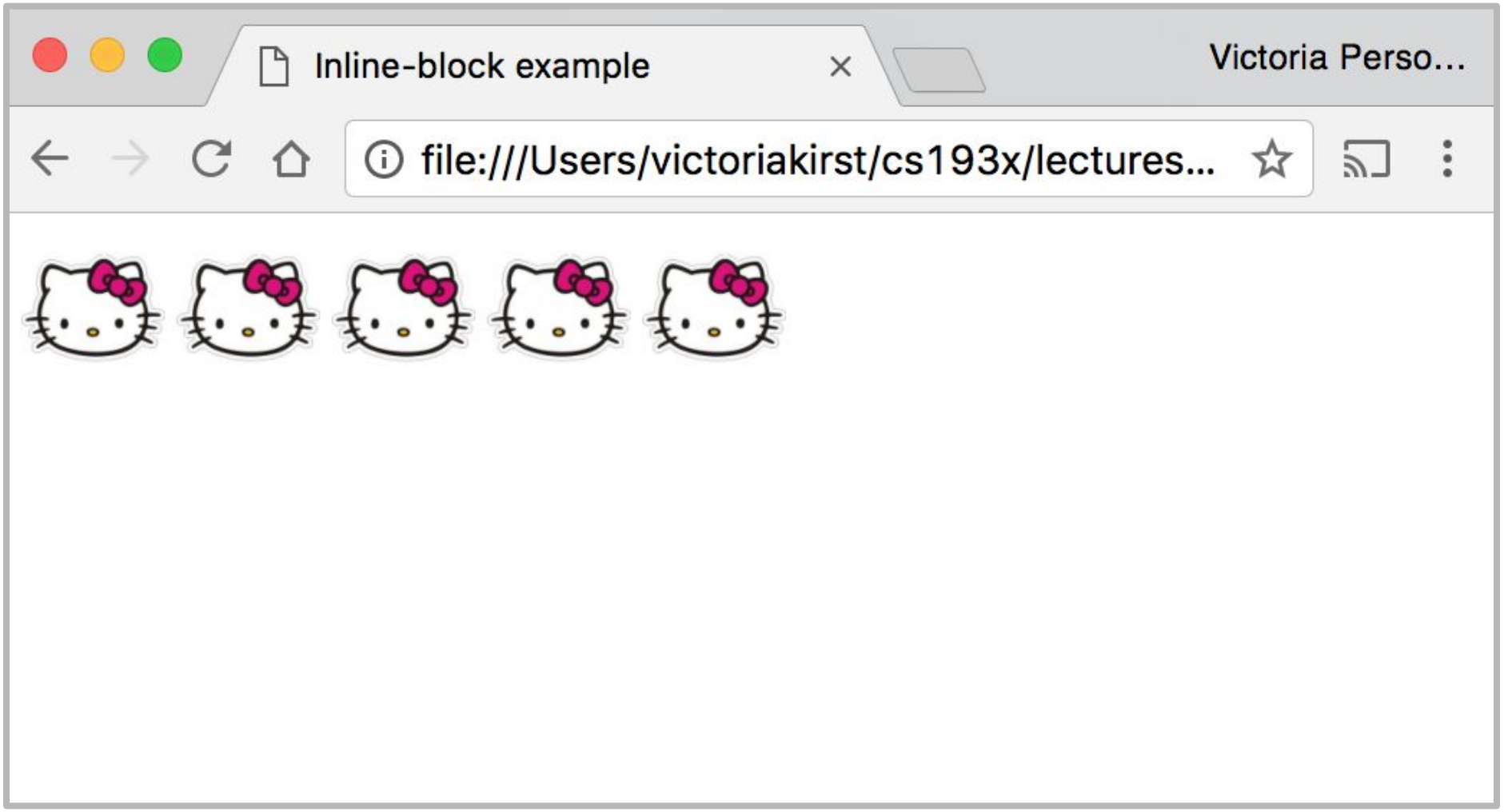
Example: Inline-block

```
img {
  width: 50px;
}
```

Q: What does this look like in the browser?

```
<img src="http://i.imgur.com/WJToVGv.jpg" />
<img src="http://i.imgur.com/WJToVGv.jpg" />
<img src="http://i.imgur.com/WJToVGv.jpg" />
<img src="http://i.imgur.com/WJToVGv.jpg" />
<img src="http://i.imgur.com/WJToVGv.jpg" />
```

http://i.imgur.com/WJToVGv.jpg =

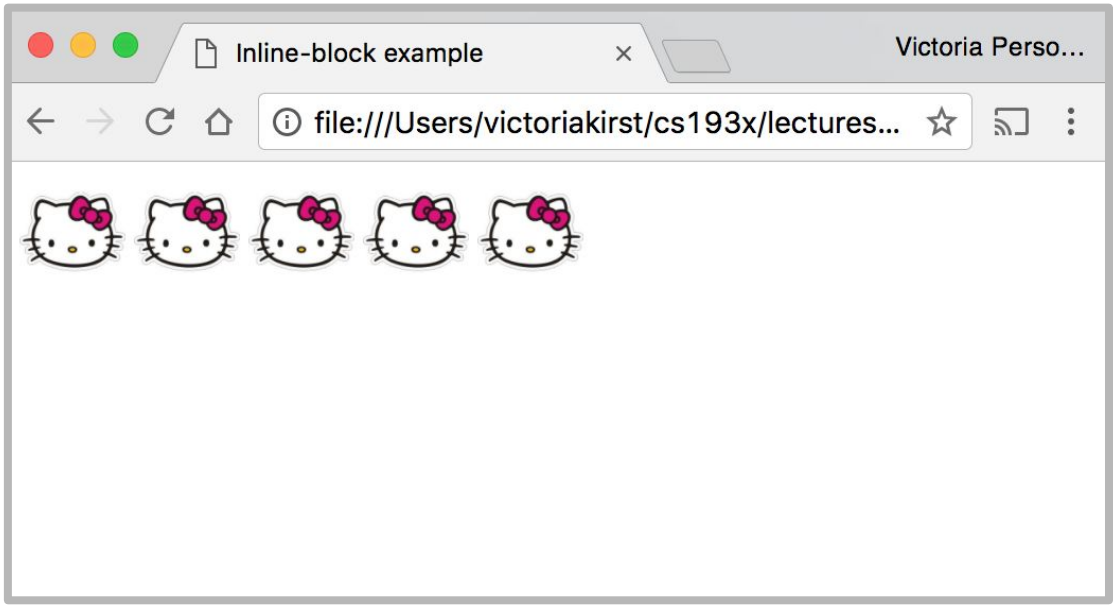

Inline-block example
Victoria Perso...
file:///Users/victoriakirst/cs193x/lectures...

Inline-block

Has width and height; flows left to right

Can set **`width`** on inline-block element, so image width is set to 50px. (JSBin)

`inline-block` flows left to right, so images are right next to each other.




```
img {
  width: 50px;
}
```

```
<img src="http://i.imgur.com/WJToVGv.jpg" />
<img src="http://i.imgur.com/WJToVGv.jpg" />
<img src="http://i.imgur.com/WJToVGv.jpg" />
<img src="http://i.imgur.com/WJToVGv.jpg" />
<img src="http://i.imgur.com/WJToVGv.jpg" />
```

The `display` CSS property

You can change an element's default rendering type by changing the **`display`** property. Examples:

```
p {
 display: inline;
}
```

```
a {
 display: block;
}
```

Possible values for `display`:

- `block`
- `inline`
- `inline-block`
- some others: link

Review

1. **block:** flows **top-to-bottom**; **has** **`height`** and **`width`**
   `<p>, <h1>, <blockquote>, <ol>, <ul>, <table>`
2. **inline:** flows **left-to-right**; **does not have** **`height`** and **`width`**
   `<a>, <em>, <strong>,<br>`
   a. **inline block:** flows **left-to-right**; **has** **`height`** and **`width`**
      `<img>`

Questions?

Moral of the story:

If your CSS isn't working, see if you're trying to apply block-level properties to inline elements